NØthing is Logical

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Slides: https://www.marialoni.org/resources/Peking26.pdf

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NØthing is logical (Nihil)

- Goal of the project: a formal account of a class of natural language inferences which deviate from classical logic
- Common assumption: these deviations are not logical mistakes, but consequence of pragmatic enrichments (Grice)
- Initial strategy: develop *logics of conversation* which model next to literal meanings also pragmatic factors and the additional inferences which arise from their interaction
- Novel hypothesis: **neglect-zero** tendency (a cognitive bias rather than a conversational principle) as crucial factor
- Main conclusion: deviations from classical logic consequence of enrichments albeit not (always) of the canonical Gricean kind
  - Dominant view: enrichments always product of **rational interactions** between cooperative language users
  - Nihil hypothesis: enrichments sometimes result from our preference to minimize cognitive effort (**bias-driven** vs product of rational choices)

Nihil team


Anttila


Degano


Klochowicz


Knudstorp


Ramotowska


Yan


Zhou

...

Non-classical inferences

Free choice (FC)

(1) FC: $\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond\alpha \wedge \Diamond\beta$ [von Wright 1968]

(2) Deontic FC inference [Kamp 1973]

a. You may go to the beach *or* to the cinema.
b. $\rightsquigarrow$ You may go to the beach *and* you may go to the cinema.

(3) Epistemic FC inference [Zimmermann 2000]

a. Mr. X might be in Victoria *or* in Brixton.
b. $\rightsquigarrow$ Mr. X might be in Victoria *and* he might be in Brixton.

Ignorance

(4) The prize is either in the garden *or* in the attic $\rightsquigarrow$ The speaker doesn't know which [Grice 1989, p.45]

(5) ? I have two *or* three children.

- In the standard approach, ignorance is a conversational implicature
- Less consensus on free choice inferences analysed as conversational implicatures; grammatical (scalar) implicatures; semantic entailments; ...

Paradox of Free Choice

Challenge 1: Explosion

- FC inference invalid in classical modal logic but robust in natural language:

  (6) a. You can have cake or ice cream $\rightsquigarrow$ You can have cake and you can have ice cream

  b. FC: $\Diamond(a \vee b) \rightsquigarrow \Diamond a \wedge \Diamond b$

- Adding FC to classical modal logic (ML) implies the equivalence of any two possibility claims [Kamp 1973]

$$\Diamond a \Rightarrow_{\textit{or}\text{-intro}+\Diamond\text{-mon}} \Diamond(a \vee b) \Rightarrow_{\text{FC}} \Diamond b$$

Challenge 2: Double Prohibition (DP)

- FC effects systematically disappear in negative contexts [Alonso-Ovalle 2006]

  (7) a. You can't have cake or ice cream $\rightsquigarrow$ You can't have cake and you can't have ice cream

  b. DP: $\neg\Diamond(a \vee b) \rightsquigarrow \neg\Diamond a \wedge \neg\Diamond b$

- The combination of FC & DP leads again to explosion [Goldstein 2019]

FC: $\Diamond(a \vee b) \models \Diamond b \Rightarrow_{\textit{contraposition}} \neg\Diamond b \models \neg\Diamond(a \vee b)$

DP: $\neg\Diamond(a \vee b) \models \neg\Diamond a \Rightarrow_{\textit{transitivity}} \neg\Diamond b \models \neg\Diamond a \Rightarrow_{\textit{contraposition}} \Diamond a \models \Diamond b$

Novel hypothesis: neglect-zero

- FC and ignorance inferences are [≠ semantic entailments]
  - Not the result of Gricean reasoning [≠ conversational implicatures]
  - Not the effect of applications of covert grammatical operators [≠ grammatical (scalar) implicatures]
- They are rather a consequence of something else speakers do in conversation, namely,

  NEGLECT-ZERO
  when interpreting a sentence speakers construct models depicting reality (some verifying the sentence, some falsifying it) ↦ common assumption
  and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*) ↦ novel hypothesis

勿空

- Tendency to neglect zero-models follows from the cognitive difficulty of:
  1. conceiving emptiness, the absence of things rather than their presence
  2. evaluating truths with respect to empty witness sets

[Nieder 2016; O. Bott *et al* 2019]

Novel hypothesis: neglect-zero

Illustration

(8) Less than three squares are black.

a. Verifier: [■, □, ■]
b. Falsifier: [■, ■, ■]
c. Zero-models: [□, □, □]; [■, ■, ■]; . . .

Zero-models in (8-c) verify the sentence by virtue of an empty set of black squares

- Cognitive difficulty of zero-models confirmed by experimental findings and connected to / can be argued to explain:
  - the special status of 0 among the natural numbers [Nieder 2016]
  - why downward-monotonic quantifiers are more costly to process than upward-monotonic ones (*less* vs *more*) [O. Bott *et al* 2019]
- NZ hypothesis: neglect-zero also at the origin of many common departures from classical reasoning
  1. FC and ignorance [MA 2022]
  2. Existential Import: every A is B ⇒ some A is B
  3. Principles from connexive logic: [Ziegler, Knudstorp & MA 2026]
     - Aristotle's Thesis: if not A then A ⇒ ⊥
     - Boethius' Thesis: if A then B & if A then not B ⇒ ⊥
  4. ESQ inferences & modified numerals [MA & vOrtmondt 2023]; Non-monotonic attitudes [Yan 2024]; SDA in counterfactuals [Klochowicz 2026], . . .

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(9) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(10) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(11) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(12) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. Zero-models: ?

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(13) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(14) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(15) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(16) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (16-c): verify the sentence by virtue of an empty witness for one of the disjuncts

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(17) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(18) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(19) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(20) M ate an apple or a banana.

a. Verifier: [🍎 | 🍌] ⇐ 'split'

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (20-c): sentence verified by virtue of an empty witness for one of the disjuncts
- **Split state** in (20-a): multiple (possibly conflicting) alternatives processed in a parallel fashion
- Neglect-zero hypothesis: Zero-models, where only one of the disjuncts is depicted, are cognitively taxing and therefore disregarded
- The exclusion of zero-models from the set of verifiers for a disjunction leads directly to an account of ignorance inferences and FC

A new conjecture: no-split

(21) Maria ate an apple or a banana.

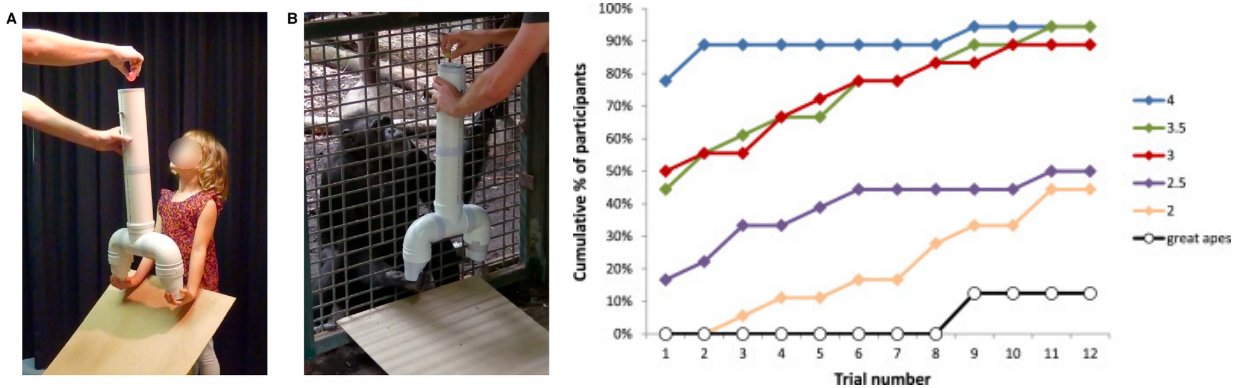
a. Verifier: [🍎 | 🍌] [⇐ **split state**]

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- **Split states**: multiple (possibly conflicting) alternatives processed in a parallel fashion ↦ also a cognitively taxing operation

NO-SPLIT CONJECTURE [Klochowicz, Sbardolini & MA, SuB 2025]
the ability to split states (entertain multiple possibilities) is developed late



Preschoolers have trouble conceiving possible different futures [Redshaw & Suddendorf 2016]

- Proposal: Combination of neglect-zero + no-split can explain conjunctive interpretations of disjunction observed in pre-school children

No-split: conjunctive children

- Children (3-5/6y) sometimes (but systematically) interpret disjunctions conjunctively [Evans 93; Singh *et al* 16 (cf. Skordos *et al* 20); Cochard25; Bleotu *et al* 25]

(22) M ate an apple *or* a banana $\rightsquigarrow$ M ate an apple *and* a banana
$\alpha \vee \beta \rightsquigarrow \alpha \wedge \beta$

(23) M *can* eat an apple *or* a banana $\rightsquigarrow$ M *can* eat an apple *and* a banana
$\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond(\alpha \wedge \beta) \not\equiv \Diamond\alpha \wedge \Diamond\beta$

(24) M did *not* eat apples *or* bananas $\rightsquigarrow$ M *neither* ate apples *nor* bananas
$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$

- Proposal: children have conjunctive readings as they (similarly to adults) neglect zero and, unlike adults, lack the ability to split

1. Deriving ignorance:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{SPLIT}}$ [🍎 | 🍌]

   $\rightsquigarrow$ It might be an apple and it might be a banana (adults)

2. Deriving conjunctive reading:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{NO-SPLIT}}$ [🍎 🍌]

   $\rightsquigarrow$ Both an apple and a banana (children)

3. In case of incompatible alternatives: [Leahy & Carey 2020, Bleotu *et al*]

   Left OR right $\Rightarrow_{\text{NZ}}$ ↙ + ↘ $\Rightarrow_{\text{NO-SPLIT}}$ **contradiction** ($\bot$)

   $\rightsquigarrow$ Sequential or first disjunctive interpretations (children)

Cognitive bias approach (NIHIL$^+$)

Common assumption: Reasoning and understanding of natural language involve the creation of mental models [Johnson-Laird 1983, Mascarenhas & Koralus 2013, a.o.]

- **Understanding** a sentence S means being able to mentally construct a model picturing the world which verifies S, and possibly also a model which falsifies it
- **Reasoning** depends on two main processes: first construct verifying models for the premises and then check the validity of the conclusion on these models

Novel hypothesis: biases can constrain the construction of these models and therefore impact both reasoning and interpretation:

- NEGLECT-ZERO prevents the constructions of zero-models;
- NO-SPLIT expresses a dispreference for split-states;
- ...

Cognitive bias approach (NIHIL$^+$)

Comparison with competing accounts

	Ignorance	**FC & DIST**	**ES-Quant**	**Scalar impl.**	**Conj *or* & homog.**
Neo-Gricean	reasoning	reasoning	reasoning	reasoning	—
Grammatical	debated	grammar	grammar	grammar	grammar
NIHIL$^+$	neglect-zero	neglect-zero	neglect-zero	—	negl-z + no-split

- Standard ALT-based derivation of **scalar implicatures** [Grice 75, Horn 72]

  (25) Maria ate an apple or a banana $\rightsquigarrow$ not both
  - a. Literal meaning: $\alpha \vee \beta$ [inclusive]
  - b. ALT: $\alpha \wedge \beta$
  - c. Negation of ALT: $\neg(\alpha \wedge \beta)$ [exclusive]

- NIHIL$^+$: **conversational inferences** sometimes bias-driven (*fast, effortless*) rather than product of alternative-based reasoning (*slow, costly*)

NEXT

- Logical modelling of biases in team semantics
- Experimental findings
  1. Degano *et al* (Nat Lang Sem, 2025): ignorance ⇐
  2. NIHIL *et al* (SuB24-25, CogSci25, XPRAG25): scalar, DIST & ES-Quant
  3. Bleotu *et al* (TbiLLC 2025, SuB25): on conjunctive *or*

Modelling biases in team semantics

General methodology

Natural language sentences translated into classical logic formulas interpreted in a team semantics which models both classical and bias-driven interpretations

NAT LANGUAGE $\mapsto$ LOGICAL LANG

α (**classical**) $\Rightarrow$
$\Rightarrow$ Gric
$\Rightarrow$ ean
$\Rightarrow$ Prag
$[\alpha]^+$ (negl-zero) $\Rightarrow$ mat
$[\alpha]^{++}$ (no-split) $\Rightarrow$ ics
$\ldots$ $\Rightarrow$

Back to FC challenges

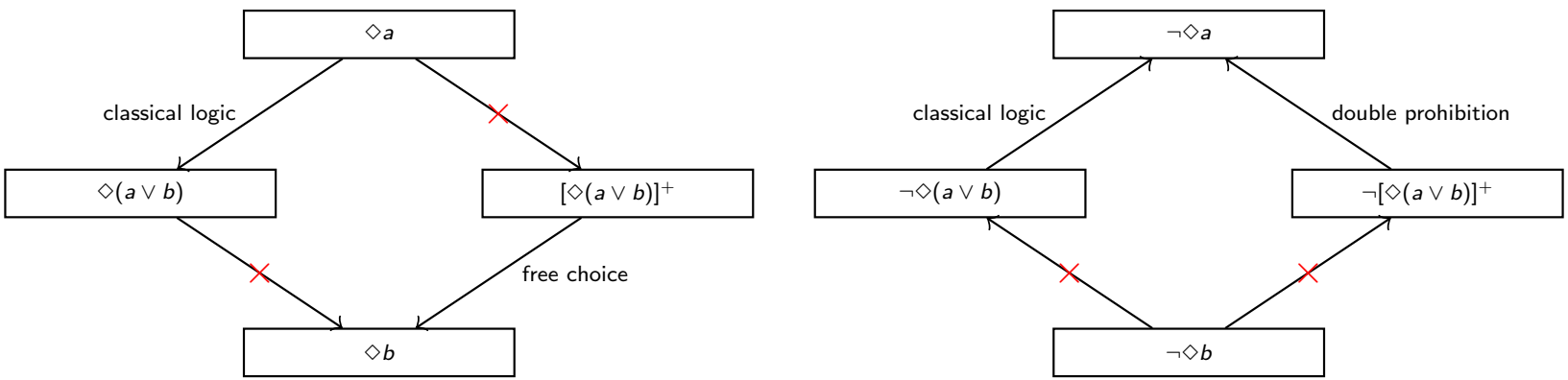


Figure: FC derived only for neglect-zero (NZ) enriched formulas; NZ enrichment vacuous under negation $\mapsto$ failure of contraposition

Modelling biases in team semantics

Team semantics

- Formulas interpreted wrt a set of points of evaluation (a team) rather than single ones [Hodges 1997; Väänänen 2007; Yang & Väänänen 2016]
  - Classical modal logic: $[M = \langle W, R, V \rangle]$

$$M, w \models \phi, \text{ where } w \in W$$

  - Team-based modal logic:

$$M, t \models \phi, \text{ where } t \subseteq W$$

- Two crucial features
  - The empty set is among the possible teams ($\emptyset \subseteq W$) $\mapsto$ zero-models
  - Multi-membered teams can model parallel processing of alternatives $\mapsto$ split states

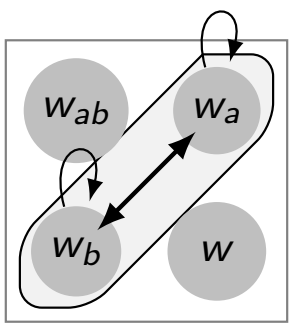


Figure: A multi-membered team modelling the parallel processing of possibilities *a* & *b*

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Neglect-zero & no-split

- Model-theoretic method: biases modelled by disallowing empty (neglect-zero) and multi-membered teams (no-split)
- Syntactic method: biases modelled via new logical atoms/operators
  - Neglect-zero: via **non-emptiness atom** NE which disallows empty teams as possible verifiers [Yang & Väänänen 2017]

$$M, t \models \text{NE} \text{ iff } t \neq \emptyset$$

  - No-split: via **flattening operator** F which induces pointwise evaluations and therefore mimics ban on parallel processing of alternatives

$$M, t \models \text{F}\phi \text{ iff for all } w \in t : M, \{w\} \models \phi$$

Modelling biases in team semantics

Classical and enriched interpretations

- $\alpha \Rightarrow$ empty team & split allowed $\mapsto$ **classical (literal) meaning**
- $[\alpha]^+ \Rightarrow$ empty team not allowed $\mapsto$ **NZ-enriched**
- $[\alpha]^{++} \Rightarrow$ neither empty team nor split allowed $\mapsto$ **NZ+NS-enriched**

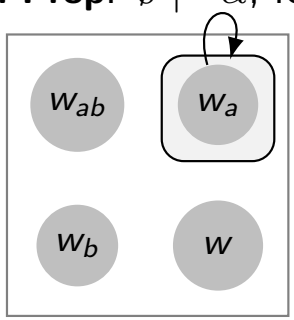
Tensor disjunction in team semantics

$$M, s \models \phi \vee \psi \text{ iff there are } t, t' : t \cup t' = s \;\&\; M, t \models \phi \;\&\; M, t' \models \psi$$

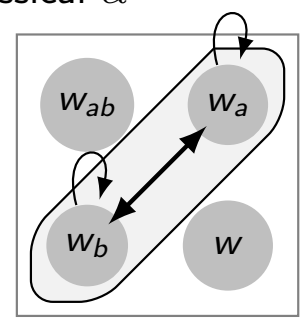
- Team s supports a **disjunction** iff s is the union of two subteams, each supporting one of the disjuncts
  - $a \vee b$: empty team allowed $\mapsto$ subteams can be empty (**classical**)
  - $[a \vee b]^+$: empty team not allowed $\mapsto$ subteams cannot be empty (**NZ-enr**)
  - $[a \vee b]^{++}$: neither empty team nor split allowed $\mapsto$ no empty or distinct subteams allowed (**NZ+NS-enriched**)

Empty Team Prop: $\emptyset \models \alpha$, for all classical α

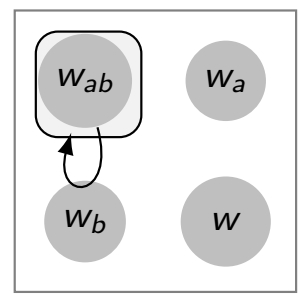
[**PROP**$= \{a, b\}$; $W = \{w_{ab}, w_a, w_b, w\}$]



$\models a \vee b$
$\not\models [a \vee b]^+$
$\not\models [a \vee b]^{++}$
$\mapsto$ **Zero-model**



$\models a \vee b$
$\models [a \vee b]^+$
$\not\models [a \vee b]^{++}$
$\mapsto$ **No-zero**, **split**



$\models a \vee b$
$\models [a \vee b]^+$
$\models [a \vee b]^{++}$
$\mapsto$ **No-zero**, **no-split**

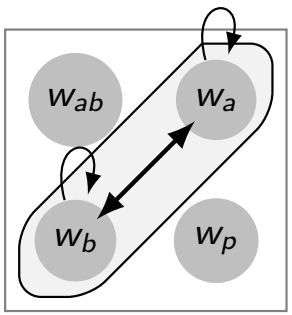
Modelling biases in team semantics: ignorance & conjunctive 'or'

(26) Maria ate an apple or a banana. $[a \vee b]$

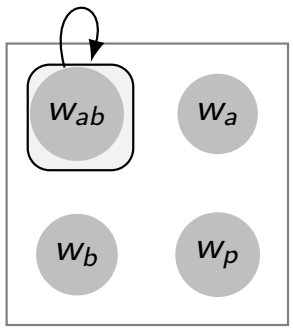
a. **No-zero**, **split** verifier: [🍎 | 🍌] $\rightsquigarrow$ it might be apple, it might be banana

b. **No-zero**, **no-split** verifier: [🍎 🍌] $\rightsquigarrow$ both apple and banana

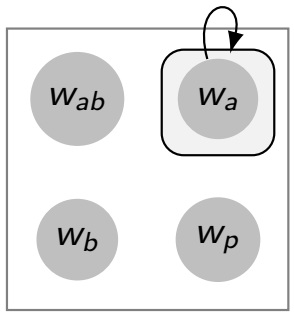
c. **Zero-models**: [🍎] ; [🍌] d. Falsifier: [🍐] $\rightsquigarrow$ neither apple nor banana



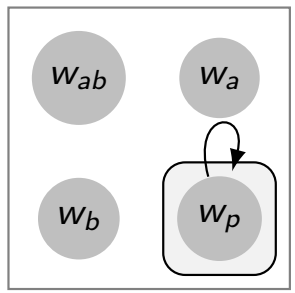
No-zero, **split** verifier



No-zero, **no-split** verifier



Zero-model



Falsifier

Predictions

We derive ignorance & conjunctive 'or' interpretations for nz- and ns-enriched formulas resp. with no undesirable effects under negation

(27) M ate an apple or a banana $\rightsquigarrow_{nz}$ It might be an apple and it might be a banana

$[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta$ (where R is state-based)

(28) M ate an apple or a banana $\rightsquigarrow_{ns}$ M ate both apple and banana

$[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$

(29) M didn't eat an apple or a banana $\rightsquigarrow_{(nz/ns)}$ M ate neither apple nor banana

$[\neg(\alpha \vee \beta)]^{(++)} \models \neg\alpha \wedge \neg\beta$

and more predictions including FC, homogeneity, ...

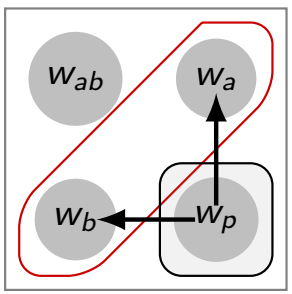
Modelling biases in team semantics: free choice

(30) Maria may eat an apple or a banana. $[\Diamond(a \vee b)]$

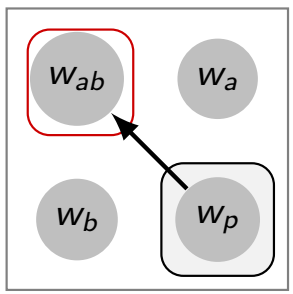
a. **No-zero**, **split** verifier: [🍎 | 🍌] $\rightsquigarrow$ may eat apple, may eat banana

b. **No-zero**, **no-split** verifier: [🍎 🍌] $\rightsquigarrow$ may eat both apple and banana

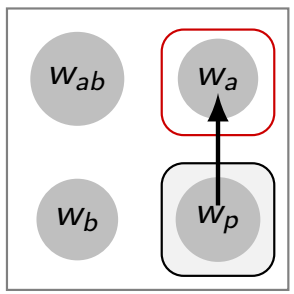
c. **Zero-models**: [🍎] ; [🍌] d. Falsifier: [🍐] $\rightsquigarrow$ may eat neither



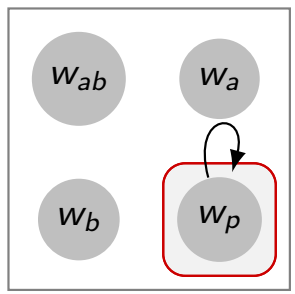
No-zero, **split** verifier



No-zero, **no-split** verifier



Zero-model



Falsifier

Predictions

We derive free choice & conjunctive permission interpretations for nz- and ns-enriched formulas resp. with no undesirable effects under negation

(31) M may eat an apple or a banana $\rightsquigarrow_{nz}$ It may be an apple and it may be a banana

$[\Diamond(\alpha \vee \beta)]^+ \models \Diamond\alpha \wedge \Diamond\beta$

(32) M may eat an apple or a banana $\rightsquigarrow_{ns}$ M may eat both apple and banana

$[\Diamond(\alpha \vee \beta)]^{++} \models \Diamond(\alpha \wedge \beta)$

(33) M may not eat an apple or a banana $\rightsquigarrow_{(nz/ns)}$ M may eat neither apple nor banana

$[\neg\Diamond(\alpha \vee \beta)]^{(++)} \models \neg\Diamond\alpha \wedge \neg\Diamond\beta$

$\Rightarrow$ Freedom of choice requires ability to conceive possible alternative futures

A closer look at these predictions

Many enrichments triggered by disjunction

(34) Maria ate an apple or a banana $(\alpha \vee \beta) \rightsquigarrow$

a. Ignorance: speaker doesn't know which
b. Conjunctive interpretation: both $(\alpha \wedge \beta)$
c. Exclusive interpretation: not both $\neg(\alpha \wedge \beta)$

Can we capture all these enrichments as nz- or ns-effects?

Two components of full ignorance: possibility vs uncertainty

(35) Maria ate an apple or a banana $\rightsquigarrow$ speaker doesn't know which

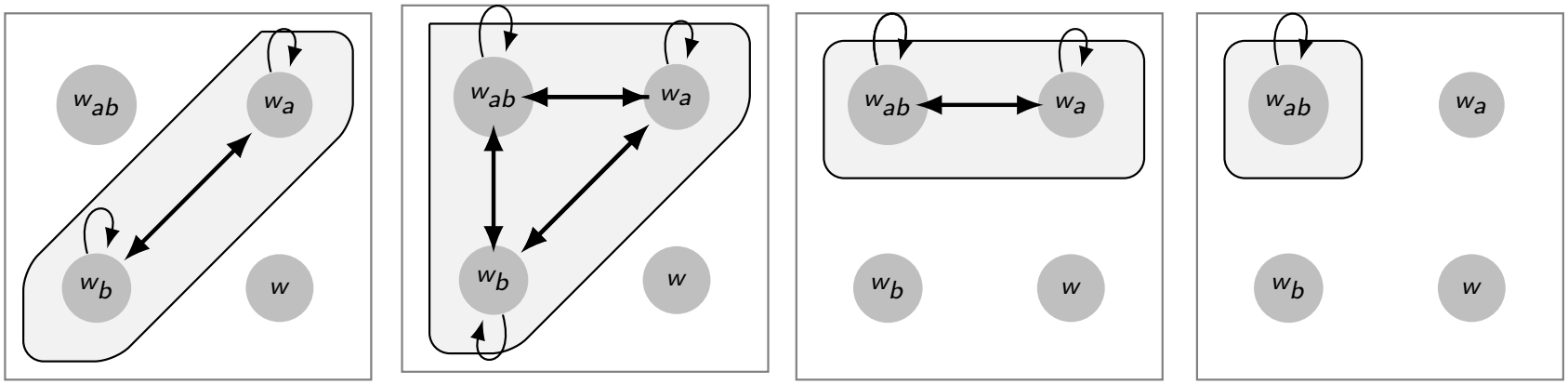
[Degano *et al* 2025][1]

a. Possibility: It is possible that M ate an apple and it is possible that M ate a banana $\Diamond_e\alpha \wedge \Diamond_e\beta$
b. Uncertainty: It is uncertain that M ate an apple and it is uncertain that M ate a banana $\neg\Box_e\alpha \wedge \neg\Box_e\beta$

[1] Degano, Marty, Ramotowska, MA, Sudo, Romoli, Breheny. The ups and downs of ignorance. *Natural Language Semantics* (2025)

Neglect-zero effects on disjunction: predictions of $[\ ]^{+\cdots}$-enrichment

Many no-zero verifiers for nz-enriched disjunction



(a) no-zero & excl $\models \neg(a \wedge b)$ (b) no-z & non-excl $\not\models \neg(a \wedge b)$ (c) no-z & no-uncertain $\not\models \neg\Box_e a$ (d) no-zero & no-split $\models (a \wedge b)$

Figure: Models for enriched $[a \vee b]^+$.

1. Neglect-zero enrichment derives **possibility**: $[\alpha \vee \beta]^+ \models \Diamond_e\alpha \wedge \Diamond_e\beta$
2. Neglect-zero + no-split derives conjunctive *or*: $[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$
3. Neglect-zero does not derive exclusive readings: $[\alpha \vee \beta]^+ \not\models \neg(\alpha \wedge \beta)$
4. Neglect-zero does not derive uncertain inferences: $[\alpha \vee \beta]^+ \not\models \neg\Box_e\alpha$

   $\mapsto$ in contrast to standard neo-Gricean approach to ignorance $\Leftarrow$

Two derivations of full ignorance

1. **Standard neo-Gricean derivation** [Sauerland 2004]

   (i) Uncertainty derived through quantity reasoning

   (36) $\alpha \vee \beta$ ASSERTION

   (37) $\neg\Box_e\alpha \wedge \neg\Box_e\beta$ UNCERTAINTY (from QUANTITY)

   (ii) Possibility derived from uncertainty and quality about assertion

   (38) $\Box_e(\alpha \vee \beta)$ QUALITY ABOUT ASSERTION

   (39) $\Rightarrow \Diamond_e\alpha \wedge \Diamond_e\beta$ POSSIBILITY

2. **Neglect-zero derivation**

   (i) Possibility derived as neglect-zero effect

   (40) $\alpha \vee \beta$ ASSERTION

   (41) $\Diamond_e\alpha \wedge \Diamond_e\beta$ POSSIBILITY (from NEGLECT-ZERO)

   (ii) Uncertainty derived from possibility and scalar reasoning

   (42) $\neg(\alpha \wedge \beta)$ SCALAR IMPLICATURE

   (43) $\Rightarrow \neg\Box_e\alpha \wedge \neg\Box_e\beta$ UNCERTAINTY

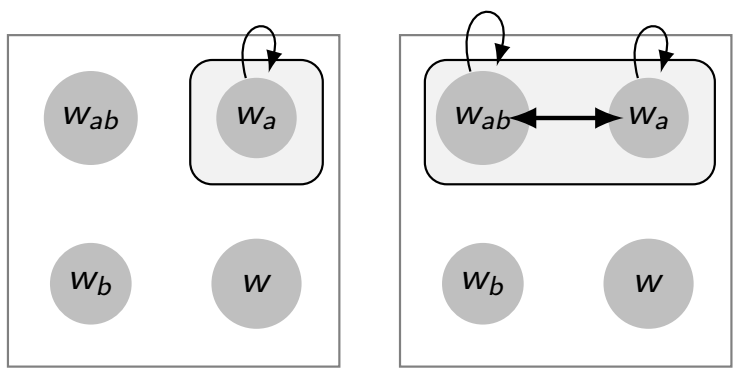
The ups and downs of ignorance [Degano et al 2025]

Contrasting predictions of competing accounts of ignorance

- Neo-Gricean: No possibility without uncertainty
- Neglect-zero: Possibility derived independently from uncertainty

Experimental findings

- Using adapted mystery box paradigm, compared conditions in which
  - both uncertainty and possibility are false [**zero-model**]
  - uncertainty false but possibility true [no-zero, no-uncertain model]
- Zero-models far more rejected (66%) than no-uncertain model (5%)
  $\Rightarrow$ Evidence that possibility can arise without uncertainty
- A challenge for the traditional neo-Gricean approach



(a) **zero-model** (b) no-uncertain

Figure: Models for $(a \vee b)$

Conclusions

- FC, ignorance inference: a mismatch between logic and language
- Grice's insight:
  - stronger meanings can be derived paying more "attention to the nature and importance to the conditions governing conversation"
- Nihil hypothesis: some strengthenings due to cognitive tendencies rather than rational choices
  - FC, possibility and related inferences as neglect-zero effects

    Literal meanings (classical fragment) + cognitive factor (NZ) $\Rightarrow$ FC, possibility, connexive principles, etc

  - Conjunctive *or*, homogeneity as no-zero + no-split effect

    Literal meanings (classical fragment) + cognitive factors (NZ, NS) $\Rightarrow$ conjunctive *or*, homogeneity

- Implementation in team-based modal logics
- Ups and downs of ignorance: a challenge for the neo-Gricean approach

THANK YOU![2]

[2]This work is supported by NWO OC project *Nothing is Logical* (grant no 406.21.CTW.023).

Appendix

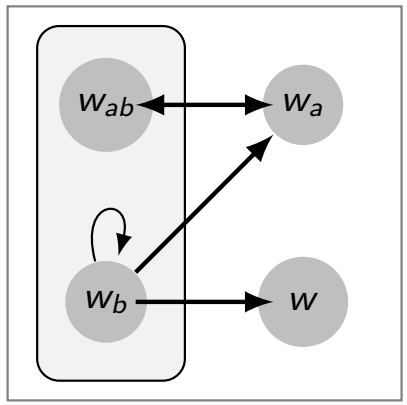
BSML: Classical Modal Logic + NE

Language

$$\phi := p \mid \neg\phi \mid \phi \vee \phi \mid \phi \wedge \phi \mid \Diamond\phi \mid \text{NE}$$

Bilateral team semantics

Given Kripke model $M = \langle W, R, V\rangle$ & teams/states $s, t, t' \subseteq W$



$$\begin{array}{rll}
M, s \models p & \text{iff} & \text{for all } w \in s : V(w, p) = 1 \\
M, s =\!| p & \text{iff} & \text{for all } w \in s : V(w, p) = 0 \\
M, s \models \neg\phi & \text{iff} & M, s =\!| \phi \\
M, s =\!| \neg\phi & \text{iff} & M, s \models \phi \\
M, s \models \phi \vee \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \;\&\; M, t \models \phi \;\&\; M, t' \models \psi \qquad \Leftarrow \\
M, s =\!| \phi \vee \psi & \text{iff} & M, s =\!| \phi \;\&\; M, s =\!| \psi \\
M, s \models \phi \wedge \psi & \text{iff} & M, s \models \phi \;\&\; M, s \models \psi \\
M, s =\!| \phi \wedge \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \;\&\; M, t =\!| \phi \;\&\; M, t' =\!| \psi \\
M, s \models \Diamond\phi & \text{iff} & \text{for all } w \in s : \text{there is } t \subseteq R[w] : t \neq \emptyset \;\&\; M, t \models \phi \\
M, s =\!| \Diamond\phi & \text{iff} & \text{for all } w \in s : M, R[w] =\!| \phi \qquad [\text{where } R[w] = \{v \in W \mid wRv\}] \\
M, s \models \text{NE} & \text{iff} & s \neq \emptyset \\
M, s =\!| \text{NE} & \text{iff} & s = \emptyset
\end{array}$$

Illustrations $\{w_{ab}, w_b\} \not\models a$; $\not\models \neg a$, but $\models a \vee \neg a$; $\{w_{ab}, w_b\} \models b$; but $\not\models \Diamond b$; $\models \Diamond a$;
$\emptyset \models a$; $\emptyset \models b$; $\emptyset =\!| a$; $\emptyset =\!| b$; $\emptyset \models \Diamond a$; $\emptyset \models \Diamond b$; $\emptyset =\!| \Diamond a$; $\emptyset =\!| \Diamond b$; $\emptyset \not\models \text{NE}$

Fact (**Empty team property**). Every NE-free (classical) formula α in BSML is supported & anti-supported by the empty team: $M, \emptyset \models \alpha$ & $M, \emptyset =\!| \alpha$

BSML: some results

Proof Theory: BSML [MA, Anttila & Yang, 2024]
`https://doi.org/10.1215/00294527-2024-0027`

Expressive completeness: BSML exp. complete for the class of all convex union-closed modal properties invariant under bounded bisimulation [Anttila & Knudstorp 25]
`https://arxiv.org/abs/2503.21850`

Comparisons via translation into MIL: BSL$_\blacklozenge$ [Knudstorp *et al* 2026]
`https://arxiv.org/abs/2610.15680`

Proof Theory: BSL$^*_\blacklozenge$ [Menorello 2026]
`https://www.illc.uva.nl/Research/Publications/Reports/MoL/`



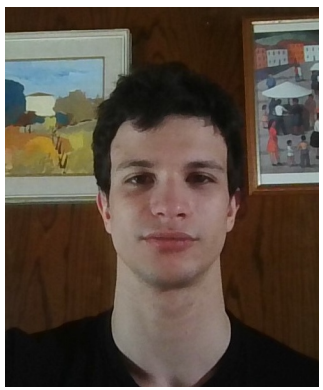
Aleksi Anttila



Søren Knudstorp



Fan Yang



E. Menorello

Implementations: two neglect-zero enrichment functions

- Two ways to model neglect-zero in BSML: [MA 2022]
  1. $[\,]^+$: enriches formulas with the requirement to satisfy NE distributed along each of their subformulas $\mapsto$ BSML$^+$
  2. $[\,]^*$: equivalent to ruling out $\emptyset$ from the set of possible teams $\mapsto$ BSML*

- For NE-free α,
  1. $[\alpha]^+$ defined as follows: [MA 2022]

$$\begin{aligned}
[p]^+ &= p \wedge \mathrm{NE} \\
[\neg\alpha]^+ &= \neg[\alpha]^+ \wedge \mathrm{NE} \\
[\alpha \vee \beta]^+ &= ([\alpha]^+ \vee [\beta]^+) \wedge \mathrm{NE} \\
[\alpha \wedge \beta]^+ &= ([\alpha]^+ \wedge [\beta]^+) \wedge \mathrm{NE} \\
[\Diamond\alpha]^+ &= \Diamond[\alpha]^+ \wedge \mathrm{NE}
\end{aligned}$$

  2. $[\alpha]^*$ defined as follows: [Klochowicz 2026]

$$\begin{aligned}
[p]^* &= (p \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\neg\alpha]^* &= \neg([\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \vee \beta]^* &= (([\alpha]^* \vee [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \wedge \beta]^* &= (([\alpha]^* \wedge [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\Diamond\alpha]^* &= (\Diamond[\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE}
\end{aligned}$$

[*-implementation $\equiv$ ruling out $\emptyset$ from the set of possible teams]

Implementations: two neglect-zero enrichment functions

- Both implementations derive possibility & free choice inferences:
  - Possibility: $[\alpha \vee \beta]^{+/*} \models \Diamond_e \alpha \wedge \Diamond_e \beta$ [R is state-based]
  - Free choice: $[\Diamond(\alpha \vee \beta)]^{+/*} \models \Diamond\alpha \wedge \Diamond\beta$
- But also strong 'neither' readings under negation:
  - Not or: $[\neg(\alpha \vee \beta)]^{+/*} \models \neg\alpha \wedge \neg\beta$
  - Double Prohibition: $[\neg\Diamond(\alpha \vee \beta)]^{+/*} \models \neg\Diamond\alpha \wedge \neg\Diamond\beta$
    [Challenge 2: failure of contraposition]
- But still differences under negation:

$\neg[p]^{+} = \neg(p \wedge \mathrm{NE}) = \neg p \vee \neg\mathrm{NE} = \neg p \vee \bot = \neg p$

$\neg[p]^{*} = \neg((p \wedge \mathrm{NE}) \vee \neg\mathrm{NE}) = \neg(p \wedge \mathrm{NE}) \wedge \neg\neg\mathrm{NE} = (\neg p \vee \neg\mathrm{NE}) \wedge \mathrm{NE} = \neg p \wedge \mathrm{NE}$

- Only BSML* derives negative possibility and negative free choice:
  - Negative possibility: $[\neg(\alpha \wedge \beta)]^{*} \models \Diamond_e\neg\alpha \wedge \Diamond_e\neg\beta$ [R is state-based]
  - Negative FC: $[\neg\Box(\alpha \wedge \beta)]^{*} \models \Diamond\neg\alpha \wedge \Diamond\neg\beta$

Implementations: adding no-split

- Implementation of no-split: uses flattening operator F

$$M, s \models \mathrm{F}\phi \text{ iff for all } w \in s : M, \{w\} \models \phi$$

$$M, s \mathrel{=\!\!|} \mathrm{F}\phi \text{ iff for all } w \in s : M, \{w\} \mathrel{=\!\!|} \phi$$

  Flattening $\mapsto$ formulas always interpreted wrt to singleton states

- Facts:
  1. OR $\Rightarrow$ AND: $\mathrm{F}[\alpha \vee \beta]^{+/*} \models \alpha \wedge \beta$
  2. Conjunctive FC: $\Diamond\mathrm{F}[\alpha \vee \beta]^{+/*} \models \Diamond(\alpha \wedge \beta)$ $\quad [\not\models \Box(\alpha \wedge \beta)]$
  3. NOT OR: $\neg\mathrm{F}[\alpha \vee \beta]^{+/*} \models \neg\alpha \wedge \neg\beta$
  4. NOT AND: $\neg\mathrm{F}[\alpha \wedge \beta]^{*} \models \neg\alpha \wedge \neg\beta$, but $\neg\mathrm{F}[\alpha \wedge \beta]^{+} \not\models \neg\alpha \wedge \neg\beta$
- Both BSML$^+$ and BSML* derive:
  - $\mapsto$ conjunctive *or* in combination with flattening (1-2)
  - $\mapsto$ classical disjunctive interpretation under negation (3)
- But only BSML* predicts
  - 'Neither' readings for NOT AND in combination with F (4) [Sbardolini 2023]
- Only in BSML$^+$, $\emptyset$ is needed as part of the building blocks

  $\Rightarrow$ Natural to assume BSML* for "conjunctive" children who plausibly do not yet have full access to $\emptyset$

Implementations: no-split enrichment

For NE- and F-free α, $[\alpha]^{**}$ defined as follows:

$$\begin{aligned}
[p]^{**} &= (\mathrm{F}p \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\neg\alpha]^{**} &= (\mathrm{F}\neg[\alpha]^{**} \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \vee \beta]^{**} &= (\mathrm{F}([\alpha]^{**} \vee [\beta]^{**}) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \wedge \beta]^{**} &= (\mathrm{F}([\alpha]^{**} \wedge [\beta]^{**}) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\Diamond\alpha]^{**} &= (\mathrm{F}\Diamond[\alpha]^{**} \wedge \mathrm{NE}) \vee \neg\mathrm{NE}
\end{aligned}$$

Predictions

1. OR $\Rightarrow$ AND: $[\alpha \vee \beta]^{**} \models \alpha \wedge \beta$
2. Conjunctive FC: $\Diamond[\alpha \vee \beta]^{**} \models \Diamond(\alpha \wedge \beta)$ $\quad [\not\models \Box(\alpha \wedge \beta)]$
3. NOT OR: $\neg[\alpha \vee \beta]^{**} \models \neg\alpha \wedge \neg\beta$
4. NOT AND: $\neg[\alpha \wedge \beta]^{**} \models \neg\alpha \wedge \neg\beta$

BSML & related systems: info states (= teams) vs possible worlds

- Failure of bivalence in BSML

$$M, s \not\models p \ \& \ M, s \not\models \neg p, \text{ for some info state } s$$

- Info states: less determinate than possible worlds
  - just like truthmakers, situations, possibilities, HYPE states, ...
- Technically:
  - Truthmakers/possibilities/HYPE states: points in a partially ordered set
  - Info states: sets of possible worlds, also elements of a partially ordered set, the Boolean lattice $Pow(W)$
- Thus systems using these structures are closely connected, although might diverge in motivation:
  - HYPE, truthmaker & possibility semantics: description of ontological structures in the world
  - BSML & inquisitive semantics: explaining patterns in inferential & communicative human activities
- NEXT:
  - Comparison via translations in Modal Information Logic [vBenthem19]

BSML & related systems: comparisons via translation

- Modal Information Logic (MIL) (van Benthem, 1989, 2019):[3] common ground where related systems can be interpreted and their connections and differences can be explored
- Goal: translations into (extensions of) MIL of the following systems:
  - Possibility semantics (Humberstone, Holliday)
  - Truthmaker semantics (Fine)
  - BSML
  - Inquisitive semantics (Ciardelli, Groenendijk & Roelofsen)
  - HYPE (Leitgeb)

  (cf. Gödel's (1933) translation of intuitionistic logic into modal logic)
- Here focus on propositional fragments:
  - disjunction
  - negation
- New result: translation of BSL (propositional fragment of BSML) in an extension of MIL

Søren Knudstorp, Maria Aloni, Alexandru Baltag, Johan van Benthem, Nick Bezhanishvili (2026). *Teams, Truthmakers, Translations*. `https://arxiv.org/abs/2610.15680`

[3] van Benthem, Implicit and Explicit Stances in Logic, *Journal of Philosophical Logic* (2019)

Modal Information Logic (MIL)

Language

$$\phi \quad ::= \quad p \mid \neg\phi \mid \phi \wedge \phi \mid \phi \vee \phi \mid \langle sup \rangle \phi\psi$$

where $p \in \mathsf{P}_\pm := \{p_+, p_- \mid p \in \mathsf{P}\}$.

Models and interpretation

Formulas are interpreted on triples $M = (X, \leq, V)$ where $\leq$ is a partial order

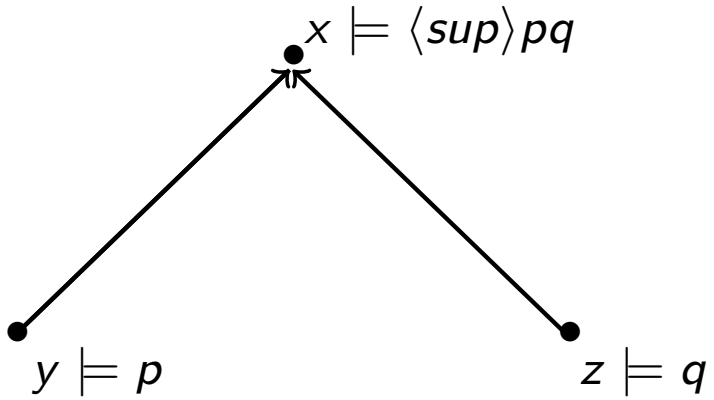
$$\begin{array}{rll}
\mathcal{M}, x \models p & \text{iff} & x \in V(p) \\
\mathcal{M}, x \models \neg\phi & \text{iff} & \mathcal{M}, x \not\models \phi \\
\mathcal{M}, x \models \phi \wedge \psi & \text{iff} & \mathcal{M}, x \models \phi \text{ and } \mathcal{M}, x \models \psi \\
\mathcal{M}, x \models \phi \vee \psi & \text{iff} & \mathcal{M}, x \models \phi \text{ or } \mathcal{M}, x \models \psi \\
\mathcal{M}, x \models \langle sup \rangle \phi\psi & \text{iff} & \text{there are } y, z : x = sup_\leq(y, z) \ \&\ \mathcal{M}, y \models \phi \ \&\ \mathcal{M}, z \models \psi
\end{array}$$

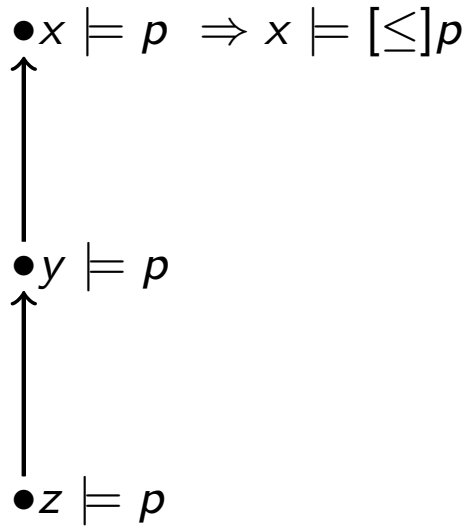
$$[\leq]\phi = \neg\langle sup \rangle(\neg\varphi)\top$$

$$\begin{array}{rll}
\mathcal{M}, x \models [\leq]\phi & \text{iff} & \text{for all } y : y \leq x \ \Rightarrow\ \mathcal{M}, y \models \phi
\end{array}$$

Modal Information Logic (MIL)

Examples





Translations into Modal Information Logic

- Possibility semantics (Humberstone, Holliday)[4]

$$\begin{aligned} tr(p) &= [\leq]\langle\leq\rangle p \\ tr(\neg\phi) &= [\leq]\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= [\leq]\langle\leq\rangle(tr(\phi) \vee tr(\psi)) \\ &\ldots \end{aligned}$$

- Truthmaker semantics (Fine)[5] (MIL on (distributive) semilattices)

$$\begin{aligned} (p)^+ &= p_+ \\ (p)^- &= p_- \\ (\neg\phi)^+ &= (\phi)^- \\ (\neg\phi)^- &= (\phi)^+ \\ (\phi \wedge \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\ (\phi \wedge \psi)^- &= (\phi)^- \vee (\psi)^- \\ (\phi \vee \psi)^+ &= (\phi)^+ \vee (\psi)^+ \\ (\phi \vee \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^- \end{aligned}$$

[4] van Benthem, Bezhanishvili, Holliday, A bimodal perspective on possibility semantics. *Journal of Logic and Computation* (2017)

[5] van Benthem, Implicit and Explicit Stances in Logic. *Journal of Philosophical Logic* (2019)

Translations into (extended) Modal Information Logic

- Truthmaker semantics (Fine) (MIL on (distributive) semilattices)

$$
\begin{aligned}
(p)^+ &= p_+ \\
(p)^- &= p_- \\
(\neg\phi)^+ &= (\phi)^- \\
(\neg\phi)^- &= (\phi)^+ \\
(\phi \wedge \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\
(\phi \wedge \psi)^- &= (\phi)^- \vee (\psi)^- \\
(\phi \vee \psi)^+ &= (\phi)^+ \vee (\psi)^+ \\
(\phi \vee \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^-
\end{aligned}
$$

- Classical fragment BSL (MIL* on distributive semilattices)

$$
\begin{aligned}
(p)^+ &= [\leq]\langle \mathrm{sup}^*\rangle p_+ \\
(p)^- &= [\leq]\langle \mathrm{sup}^*\rangle p_- \\
(\neg\phi)^+ &= (\phi)^- \\
(\neg\phi)^- &= (\phi)^+ \\
(\phi \wedge \psi)^+ &= (\phi)^+ \wedge (\psi)^+ \\
(\phi \wedge \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^- \\
(\phi \vee \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\
(\phi \vee \psi)^- &= (\phi)^- \wedge (\psi)^-
\end{aligned}
$$

$\mathcal{M}, x \models \langle \mathrm{sup}^*\rangle\phi$ iff there exist $x_1, \ldots, x_n$ s.t. each $x_i \models \varphi$ & $x = sup_{\leq}\{x_1 \cdots x_n\}$

Translations into (extended) Modal Information Logic

- Inquisitive semantics (Groenendijk, Roelofsen & Ciardelli)

$$\begin{aligned} tr(p) &= [\leq]\langle \mathrm{sup}^* \rangle p \\ tr(\neg\phi) &= [\leq]\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= tr(\phi) \vee tr(\psi) \\ tr(\phi \to \psi) &= [\leq](tr(\phi) \to tr(\psi)) \end{aligned}$$

- HYPE (Leitgeb)[6] (bimodal perspective needed for negation)

$$\begin{aligned} tr(p) &= [\leq]p \\ tr(\neg\phi) &= \blacksquare\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= tr(\phi) \vee tr(\psi) \\ tr(\phi \to \psi) &= [\leq](tr(\phi) \to tr(\psi)) \end{aligned}$$

$\mathcal{M}, x \models \blacksquare\phi$ iff for all y: $xCy \Rightarrow \mathcal{M}, y \models \varphi$

Given a compatibility model $\mathcal{M} = (X, C, \leq, V)$

[6] Odintsov, S., Wansing, H. Routley Star and Hyperintensionality. J Philos Logic (2021)

Disjunction and Negation

- Three notions of disjunction expressible in (extended) MIL:
  - Boolean disjunction: $\phi \vee \psi$
    [classical logic, intuitionistic logic, inquisitive logic]
  - Lifted/tensor/split disjunction: $\langle sup \rangle \phi\psi$
    [BSML, dependence logic, team semantics, operational semantics for Positive R]
  - Cofinal disjunction: $[co](\phi \vee \psi)$ (where $[co]\phi =: [\leq]\langle \leq \rangle \phi$)
    [possibility semantics, dynamic semantics]
- Four notions of negation:
  - Boolean negation: $\neg\phi$
    [classical logic, ...]
  - Bilateral negation: $(\neg\phi)^+ = (\phi)^- \ \& \ (\neg\phi)^- = (\phi)^+$
    [truthmaker semantics, BSML, ...]
  - Intuitionistic-like negation: $[\leq]\neg\phi$
    [possibility semantics, inquisitive semantics, intuitionistic logic]
  - Modal negation: $\blacksquare\neg\phi$
    [HYPE, ...]
- Some combinations:
  - Boolean disjunction + boolean negation $\mapsto$ classical logic
  - Boolean notions in other combinations can generate non-classicality:
    - Boolean disjunction + intuitionistic negation $\mapsto$ intuitionistic/inquisitive logic[7]
  - Classicality also generated by non-boolean combinations:
    - Split disjunction + bilateral negation (classical fragm. BSPL)
    - Cofinal disjunction and intuitionistic negation (possibility semantics)

[7] Inquisitive & intuitionistic logic: same connectives but different translations for the atoms.

Full translation for BSPL

$$
\begin{array}{rcl@{\qquad}rcl}
p^+ & := & [\leq]\langle \mathsf{sup}^* \rangle p_+ & p^- & := & [\leq]\langle \mathsf{sup}^* \rangle p_- \\
\mathrm{NE}^+ & := & \bigwedge_{p \in \mathrm{P}} \neg(p^+ \wedge p^-) & \mathrm{NE}^- & := & \bigwedge_{p \in \mathrm{P}} (p^+ \wedge p^-) \\
\bot^+ & := & \mathrm{NE}^- & \bot^- & := & \top \\
(\neg\varphi)^+ & := & \varphi^- & (\neg\varphi)^- & := & \varphi^+ \\
(\varphi \vee \psi)^+ & := & \langle \mathsf{sup} \rangle \varphi^+ \psi^+ & (\varphi \vee \psi)^- & := & \varphi^- \wedge \psi^- \\
(\varphi \wedge \psi)^+ & := & \varphi^+ \wedge \psi^+ & (\varphi \wedge \psi)^- & := & \langle \mathsf{sup} \rangle \varphi^- \psi^-
\end{array}
$$

Theorem (Knudstorp 2026)

$$\varphi \models_{BSPL} \psi \qquad \text{iff} \qquad \Gamma, \varphi^+ \models_{MIL^*} \psi^+$$

where

$$\Gamma := \left\{ [\leq](\mathrm{NE}^+ \vee \mathrm{NE}^-), \bigwedge_{p \in \mathrm{P}} \langle \mathsf{sup} \rangle p^+ p^- \right\}$$

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